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Despite our best efforts to make this thesis as complete as possible, shortcomings and limitations are inevitable. We sincerely hope to receive your understanding and constructive

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Abstract

A good abstract is a stand-alone summary of the paper, and should summarize the key components of the manuscript. Generally, the abstract should be concise and informative within 200-300 words. As an abstract is a separate section, it should be a self-containing text (no abbreviations, no references, no URLs, no undefined concepts, etc.). For research articles, abstracts should give a pertinent overview of the work. We strongly encourage authors to use the following style of structured abstracts, but without headings: (1) Background: Place the question addressed in a broad context (research background) and highlight the purpose of the study; (2) Methods: Describe briefly the main methods or treatments applied; (3) Results: Summarize the article's main findings; and (4) Conclusions: Indicate the main conclusions or interpretations.

Keywords: cybersecurity, information security, threat analysis, resilience

List of Abbreviations

Acronym	Definition
DSP	Digital Service Providers
EU	European Union
GSP	Global Stability Protection
OES	Operators of Essential Services

Chapter 1

Introduction

1.1 Background

The Internet of Things (IoT) connects physical objects, sensors, and various devices through communication networks, enabling data-driven services in domains such as healthcare, industrial automation, and smart environments. Alongside the expansion of IoT deployments, the number of connected components exposed to cybersecurity threats increases correspondingly [1]. The Message Queuing Telemetry Transport (MQTT) protocol is commonly used for lightweight publish-subscribe communication in IoT environments. However, MQTT-based systems can become targets for denial-of-service (DoS) and distributed denial-of-service (DDoS) attacks. The DoS/DDoS-MQTT-IoT dataset, collected from a physical IoT testbed, provides normal MQTT traffic alongside ten DoS attack scenarios to facilitate the study of attacks in this context [2].

Intrusion Detection Systems (IDS) monitor network activity to identify traffic or behavior exhibiting malicious signs. Machine learning and deep learning methods have been utilized and researched for this task due to their ability to learn characteristic patterns directly from network traffic. Their effectiveness depends partly on the availability and representativeness of the training data [3]. Federated Learning (FL) allows multiple clients to collaboratively train a shared model while keeping their data local; a central server distributes the global model and aggregates updates from clients across training rounds [4]. This approach is highly suitable when IoT data is distributed across multiple devices or locations, although communication and computational costs remain critical factors to consider in resource-constrained environments [5], [6]. Keeping data local also does not equate to completely eliminating privacy concerns arising from the exchange of model updates [7].

Synthetic data generation is a potential solution to supplement existing training data sources. Conditional Variational Autoencoders (CVAEs) guide the generation process based on supplementary information, such as class labels, and are capable of modeling structured data [8]. In the intrusion detection problem, generated samples can provide additional data for specific attack classes, but their quality and utility for downstream IDS must be proven through empirical evaluation. These developments set the premise for investigating privacy-preserving synthetic data generation methods for intrusion detection in MQTT networks; specific challenges and research gaps will be discussed in detail in the next section.

1.2 Problem Statement

Building an effective IDS for IoT networks faces multiple challenges related to data privacy, the representation of attack classes, and the practical costs of the model training process. These challenges are particularly evident when training data is distributed among clients and synthetic attack samples are considered as a supplementary detection solution.

The first problem relates to privacy risks during federated training. Although FL avoids the periodic transfer of raw local data, clients must still exchange model updates with the server. Previous studies have demonstrated that shared gradients can leak information about training data in certain attack scenarios [7]. Therefore, sharing model weights instead of raw data cannot be considered a comprehensive privacy protection measure. Depending on the threat model and the type of information to be protected, supplementary mechanisms may be mandatory.

The second problem concerns the representation of attack classes in the training data. Class imbalance can bias the learning process towards majority classes, making it difficult to identify classes that account for a low percentage [3]. For the DoS/DDoS detection problem in MQTT, the impact of this issue depends on the dataset version, label mapping, preprocessing pipeline, and data partitioning strategy for clients. Therefore, the class distribution of the dataset after selection and preprocessing must be empirically verified rather than assumed by default. If some attack classes lack data, synthetic samples can supplement training examples. However, merely increasing the number of samples does not prove that the generated data is useful; their actual contribution must be evaluated through data quality metrics and the detection performance of the downstream IDS model.

The third problem involves the utility and operational costs of synthetic data generation in a federated environment. Generative models require significant resources for both training and inference, while FL incurs additional communication costs and aggregation operations. Privacy-preserving mechanisms can increase computational or communication costs depending on their design and configuration. Previous studies have investigated federated tabular data synthesis and generative data augmentation for IDS using GAN-based methods [9], [10]. The applicability of a CVAE-based method for generating specific MQTT attack data in this study—including its impact on downstream detection capabilities and resource consumption—therefore remains an open question that needs empirical verification.

The literature review indicates that related components have been studied individually. HT-Fed-GAN investigates decentralized tabular data synthesis using GANs combined with Differential Privacy [9]. FedGenID applies conditional GANs in a federated environment to augment data for intrusion detection in industrial IoT and examines the impact of Differential Privacy [10]. FLCVAE-ID combines CVAE and FL for anomaly detection in an industrial control context [11], while a more recent study, F-ACVAE, explores an adaptive CVAE-based federated approach for IoT intrusion detection [12]. These studies differ significantly in datasets, generative objectives, network architectures, privacy protection mechanisms, and evaluation contexts. Within the literature reviewed for this study, no work has been found that specifically evaluates CVAE-based DoS/DDoS MQTT data generation in FL on the DoS/DDoS-MQTT-IoT dataset, while simultaneously comparing DP and SecAgg under the same experimental protocol regarding data quality, downstream IDS performance, privacy protection characteristics, and computational and communication costs. This observation is limited to the aforementioned combination of dataset, problem, and evaluation setup; it does not imply that there is no research on CVAE, FL, DP, or SecAgg individually or in other combinations.

Based on this reality, this research addresses the following problem: How to design and empirically evaluate a CVAE-based federated framework to generate synthetic MQTT attack data, while examining data utility, privacy protection characteristics, and computational and communication costs? The study investigates two configurations: CVAE + FL + DP and CVAE + FL + SecAgg, within the experimental scope defined in this thesis.

1.3 Research Objectives

The main objective of this research is to design and evaluate PP-FedData, a privacy-preserving synthetic data generation framework based on Conditional Variational Autoencoders (CVAE) and Federated Learning (FL) for DoS/DDoS attack detection in MQTT-based IoT environments. To achieve this general objective, the study defines the following specific objectives:

- Preprocess the selected DoS/DDoS-MQTT-IoT dataset and prepare it for federated training, including feature preparation, label processing, and class distribution examination. Establish a procedure to configure and optimize hyperparameters at the global server level before distributing the training configuration and required initialization parameters to clients; the configuration selection must be based on an evaluation source that does not require centralizing raw client data (e.g., a public validation set or appropriately shared aggregated metrics).
- Design two privacy-preserving configurations in the Federated Learning environment: CVAE + FL + Differential Privacy and CVAE + FL + Secure Aggregation, using shared experimental conditions to facilitate comparison.
- Implement the federated training workflow for the two configurations: clients train the CVAE locally based on the training configuration and initialization parameters distributed from the Global Server, transmit updates through the corresponding privacy-preserving mechanism, and participate in global model aggregation across training rounds.
- Utilize the trained global CVAE model to generate conditional synthetic MQTT traffic samples according to target classes, with special attention to minority attack classes if an imbalance occurs in the processed data.
- Evaluate the generated data using appropriate data quality metrics and downstream machine learning-based intrusion detection experiments. Investigate whether the synthetic samples provide useful support for attack detection, and compare the two configurations regarding data quality, IDS effectiveness, privacy characteristics, and computational or communication costs.

1.4 Significance of the Study

This study investigates the capability of CVAE-based synthetic data generation in a Federated Learning environment for IoT attack detection on the MQTT platform. The significance of the research lies in providing an in-depth empirical analysis tailored to a specific application on how to combine generative models with privacy-preserving mechanisms.

Technical and research perspective: PP-FedData builds a comprehensive pipeline encompassing data preparation, global hyperparameter optimization, client-side CVAE training, federated aggregation, and synthetic data generation. The deployment and evaluation of DP and SecAgg as two independent configurations under shared experimental conditions aim to establish a basis for analyzing their respective impacts on data quality and system costs. The study does not pre-assume which configuration is superior; trade-offs will be analyzed based on empirical results.

Practical perspective: Synthetic attack data can supplement training examples when certain classes lack sufficient samples. If the generated data retains essential traffic characteristics and improves the performance of the downstream IDS, they can contribute to overcoming data availability barriers in the research context. Measuring privacy characteristics and resource costs also provides useful information for future development steps. The study does not assume that synthetic data can entirely replace real traffic, nor that this framework can be immediately deployed on resource-constrained IoT devices.

1.5 Scope and Limitations

This research focuses on the design, implementation, and empirical evaluation of the PP-FedData framework for synthetic data generation and DoS/DDoS attack detection in MQTT-based IoT environments. The specific scope is defined as follows:

- Application domain: MQTT-based IoT network traffic, with an emphasis on DoS/DDoS attack detection.
- Dataset: The DoS/DDoS-MQTT-IoT dataset is selected as the primary data source and is preprocessed for the model and evaluation pipeline.
- Generative model: CVAE is used to generate conditional synthetic data samples for specified classes in the prepared tabular traffic data.
- Federated architecture: Employs an FL model following a server-client architecture. Global hyperparameters are configured and optimized at the server before the training configuration is transferred to clients; the evaluation source for this process must be determined such that raw client data is not centralized (e.g., a public validation set or appropriate aggregated metrics).
- Privacy-preserving configurations: Two configurations, CVAE + FL + DP and CVAE + FL + SecAgg, are independently investigated.
- Evaluation: The study examines the quality of the generated data, the performance of the downstream IDS model, along with the recorded computational and communication costs.

The findings of this study should be interpreted in light of the following limitations:

- **Dataset and protocol coverage:** The study focuses exclusively on a specific MQTT DoS/DDoS dataset. Results may not be generalizable to other IoT protocols, deployment environments, or attack types without further supplementary experiments.
- **Dependence on data and configuration:** The quality of synthetic data depends on the selected dataset, preprocessing methods, CVAE architecture, hyperparameters, client partitioning, and training procedures. Generated samples may not fully reflect all characteristics of real network traffic.
- **Privacy assumptions:** The privacy characteristics of DP depend on the specific mathematical mechanism, privacy parameters, and the definition of neighboring data. SecAgg relies on protocol assumptions and only protects individual updates during the aggregation process; it does not inherently provide differential privacy guarantees for the post-aggregation model. Therefore, the study does not claim absolute defense against all forms of privacy attacks.
- **Computational resources:** CVAE training, federated communication, and privacy-preserving operations can incur additional resource costs. The feasibility of directly deploying the entire framework on low-power IoT devices falls outside the conclusions that can be drawn without direct empirical evaluation at the device level.
- **Experimental validity:** The obtained results depend on the software, hardware, data partitioning, hyperparameter selection procedures, and evaluation metrics used. Further verification assessments on multiple datasets and actual deployment environments are required before generalizing the conclusions.

1.6 Thesis Structure

The remainder of this thesis is structured into six chapters as follows:

- **Chapter 1: Introduction:** Outlines the background, problem, objectives, significance, scope, and structure of the study.
- **Chapter 2: Literature Review:** Reviews relevant studies and identifies the research gap and the position of this study in the existing literature.
- **Chapter 3: Methodology:** Details dataset preparation, global hyperparameter optimization, CVAE-FL training with DP or SecAgg, and evaluation methods.
- **Chapter 4: Experimental and Results:** Describes the experimental setup and reports synthetic-data, IDS, and system overhead results.
- **Chapter 5: Discussion:** Interprets the findings, examines the observed trade-offs, and discusses the limitations and implications of the study.
- **Chapter 6: Conclusion and Future Work:** Summarizes the conclusions and outlines directions for further research.

Chapter 2

Literature Review

Reviews relevant existing research and demonstrates how your thesis fits into the broader academic conversation.

2.1 Review of Previous Studies

2.2 Summary of the Literature Review

2.3 Contribution of Research

Chapter 3

Methodology

Describes the research design, data collection methods, and analysis techniques.

3.1 Research Design

3.2 Data Collection Methods

3.3 Sampling Data Analysis Techniques

3.4 Limitations of the Methodology

Chapter 4

Experimental and Results

Provide a concise and precise description of the experimental results, their interpretation as well as the experimental conclusions that can be drawn. You should discuss the results and how they can be interpreted in perspective of previous studies and of the working hypotheses.

4.1 Introduction

4.2 Presentation of Data

4.3 Analysis of Results

4.4 Interpretation of Results

4.5 Comparison with Literature

4.6 Implications of the Results

Chapter 5

Discussion

The findings and their implications should be discussed in the broadest context possible and limitations of the work highlighted.

5.1 Restate the Research Problem or Objectives

5.2 Summarize Key Findings

Chapter 6

Conclusion and Future Work

The 'conclusions' are a key component of the paper. It should complement the 'abstract' and is normally used by experts to value the paper's engineering content. A conclusion is not merely a summary of the main topics covered or a re-statement of your research problem, but a synthesis of key points and, if applicable, where you recommend new areas for future research.

6.1 Conclusion

6.2 Future Work

Appendix A

If Any

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